Name: Nathan Li

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Problem set number: 3

Due date: 2/24/2026

Acknowledgement:

“Help from Course lecture notes”

1) <u>Known</u>: long rod radius R, vol. heat gen. $\dot{q}$

$T_\infty = T_\infty$, $h = h$, $T = T(x,t)$

$k = k$, $\rho = \rho$, $c_p = c_p$

<u>Find</u>: a) energy balance

b) temp. governing diff. eq'

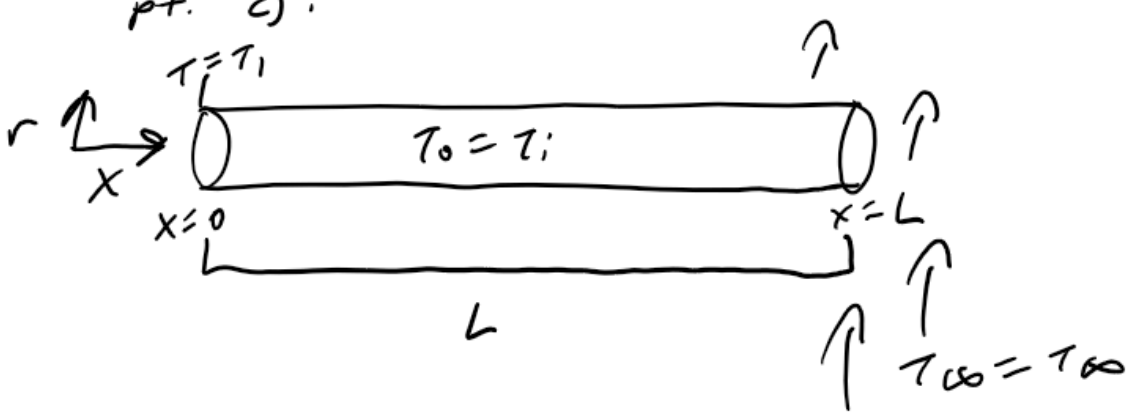
c) B.C.s & I.C.s for rod $x=0$ to $x=L$,

$T_i = T_i$

<u>Schematic</u>:

see given for a) & b),

pt. c):



<u>Assumptions</u>:

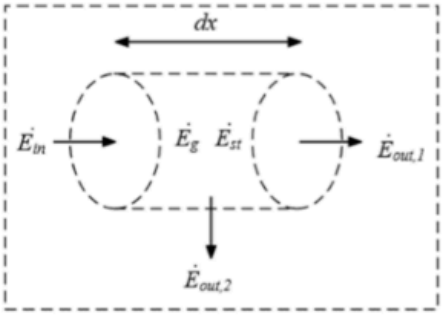
rod small radius, steady state

perfect h.t., 1D h.t.

no q radiation

Analysis:

a)



$$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out_1} - \dot{E}_{out_2} = \dot{E}_{st}$$

$$\left(-kA\frac{\partial T}{\partial x}\right) + \dot{q}A - kA\frac{\partial T}{\partial x} - hA\left(T(x,t) - T_\infty\right) = \rho c_p \frac{\partial T}{\partial t}$$

$$\boxed{kA\frac{\partial^2 T}{\partial x^2} + \dot{q}A - hA\left(T(x,t) - T_\infty\right) = \rho c_p A\frac{\partial T}{\partial t}}$$

b)

$$\boxed{k\frac{\partial^2 T}{\partial x^2} + \dot{q} - h\left(T(x,t) - T_\infty\right) = \rho c_p \frac{\partial T}{\partial t}}$$

c) $T(x,t)$

$$\boxed{T(0,t) = T_1 \quad , \quad T(x,0) = T_i}$$

$$\boxed{-k\frac{\partial T}{\partial x}\bigg|_{x=L} = h\left(T(L,t) - T_\infty\right)}$$

2) <u>Known</u>: wall A $L_A = 30\,cm$ Wall B: $L_B = 10\,cm$

$\dot{q} = \dot{q}$ $\dot{q} = 0$

$k = k_A$ $k_B = 30\ W/(m \cdot K)$

$T_\infty = 20°C$, $h = 500\ W/(m^2 \cdot K)$

$T_w = 115°C$, $T_i = 100°C$

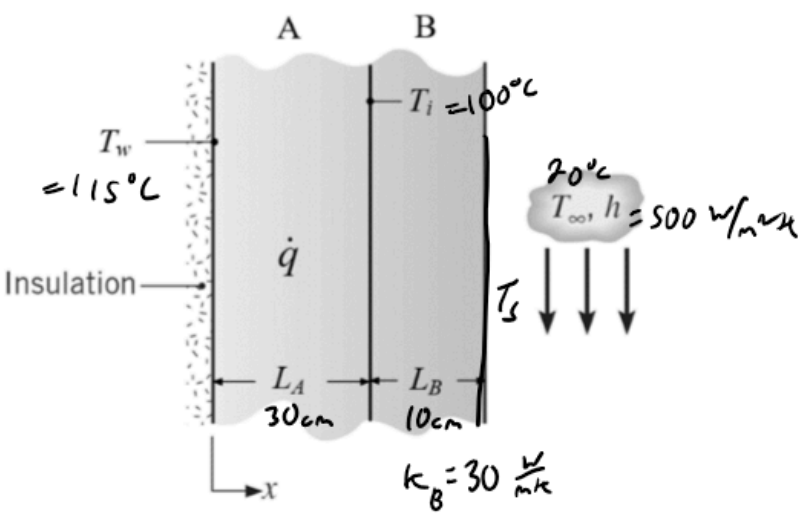
<u>Find</u>: a) right surface temp

b) uniform $\dot{q}_{st}$

c) k_A

d) T profile plot

<u>Schematic</u>:



<u>Assumptions</u>:

perfect insulation next to wall A, steady state

conduction bet A & B, 1D H.T.

convection bet. B & T_∞

no radiation

Analysis:

a) q'' thru wall $= q''$ surface

$$-k\frac{\partial T}{\partial x} = h(T_s - T_\infty)$$

$$k_b\frac{(T_i - T_s)}{L_B} = h_\infty(T_s - T_\infty)$$

$$30\frac{(100 - T_s)}{0.1m} = 500(T_s - 20)$$

$$30000 - 300T_s = 500T_s - 10000$$

$$40000 = 800T_s$$

$$\boxed{T_s = 50°C}$$

b) $q''_{cond} = q''_{conv}$

$$\dot{q}L_A = h\Delta T$$

$$\dot{q} = \frac{h\Delta T}{L_A}$$

$$= \frac{500(50-20)}{0.3m}$$

$$\boxed{\dot{q} = 50\ kW/m^3}$$

c) from LS

$$T(x) = -\frac{\dot{q}}{2k_A}x^2 + ax + b$$

B.C. $\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0, \therefore a = 0, \; b = T_w$
$= 115°C$

$T(L_A) = T_i = 100°C$

$$100 = -\frac{(50e3)}{2k_A}(0.3)^2 + 115$$

$$-15 = -\frac{2250}{k_A}$$

$$\boxed{k_A = 150 \frac{W}{mK}}$$

d) $T(x) = -\frac{\dot{q}}{2k_A}x^2 + 115$ for $0 < x < L_A$

$$= -166.67x^2 + 115$$

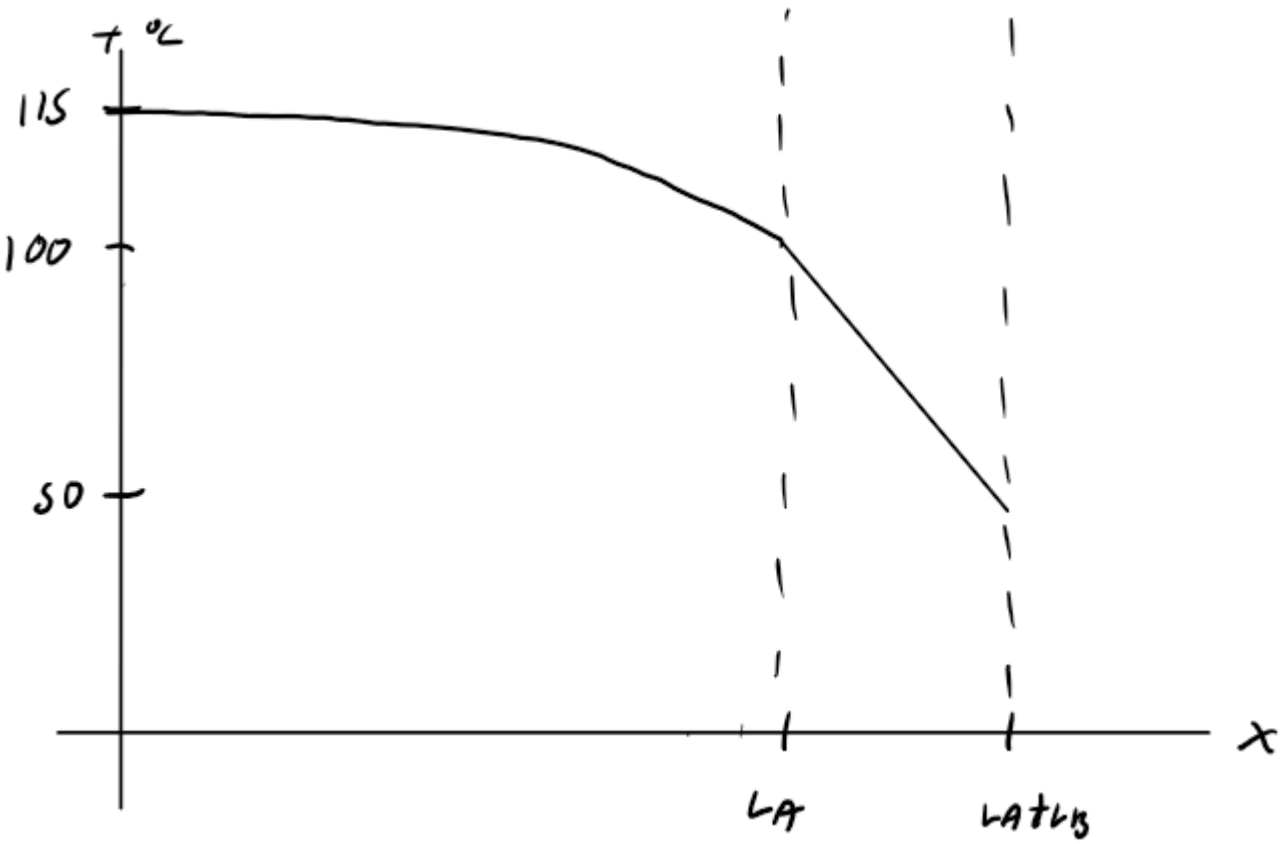
$T(L_B) = a(x - L_A) + b$

$T(L_A) = b = T_i = 100°C$

$T(L_A + L_B) = a(0.1m) + 100 = 50 = T_s$

$a = -500$

$T(x) = -500(x - 0.3) + 100$ for $L_A < x < L_B$


t °C
115
100
50
x
LA
LA+LB

3) <u>known</u>: $k_b = 25 \frac{W}{m \cdot K}$, $k_{TBC} = 1 \frac{W}{m \cdot K}$, $T_{\infty_o} = 1616K$

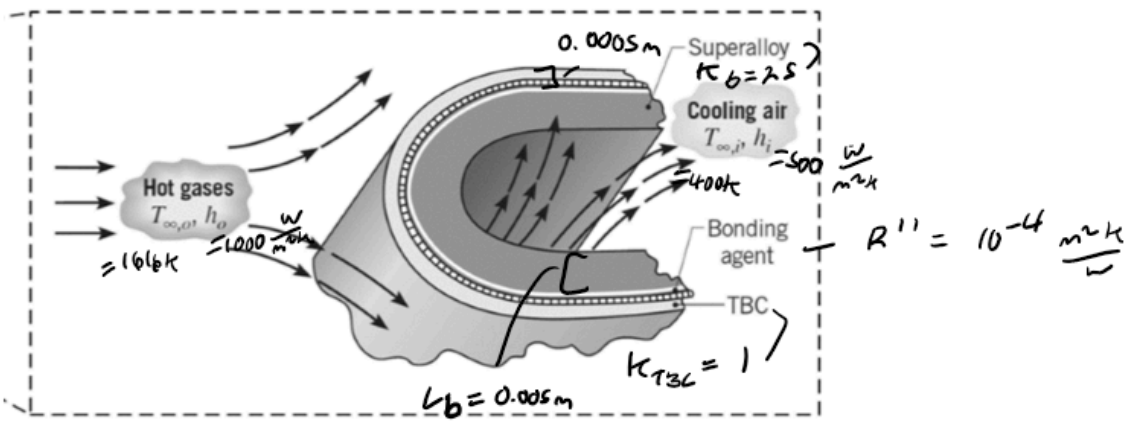
TBC bonding $R'' = 10^{-4} \frac{m^2 K}{W}$, cooling air $T_{\infty_i} = 400K$

$h_o = 1000 \, W/(m^2 \cdot K)$, $h_i = 500 \, W/(m^2 K)$, $L_b = 5mm$, $T_{max} = 1200K$, $L_{TBC} = 0.5mm$

<u>Find</u>: a) No TBC ext surface temp

b) TBC applied surface temp vs. T_{max}

<u>Schematic</u>:



<u>Assumptions</u>:

1D H.T., steady state

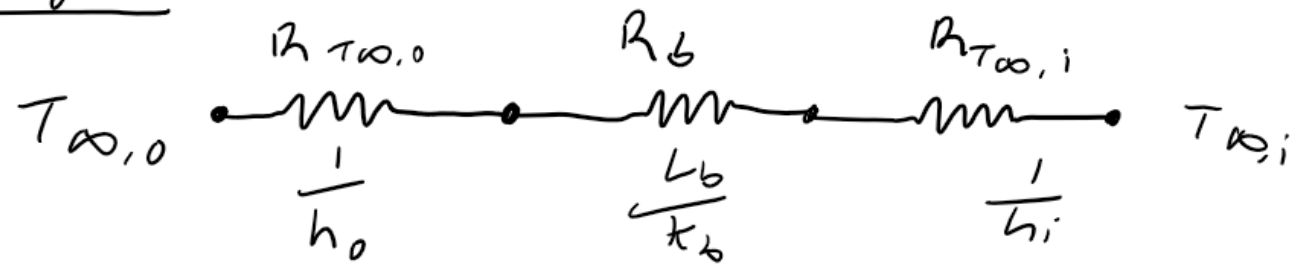
thin bonding agent

no radiation

perfect hit.

<u>Analysis</u>:

a)



$$R_{tot} = \frac{1}{1000} + \frac{0.005}{25} + \frac{1}{500} = 0.0032$$

$$q'' = \frac{\Delta T}{R_{tot}} = \frac{1616 - 400}{0.0032} = 380 \text{ kW/m}^2$$

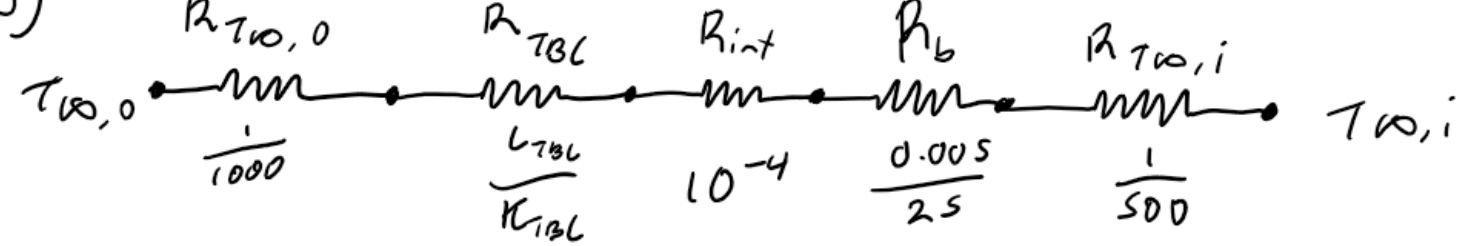
$$= h_o (T_{\infty,o} - T_s)$$

$$380e3 = 1000(1616 - T_s)$$

$$1000 T_s = 1236e3$$

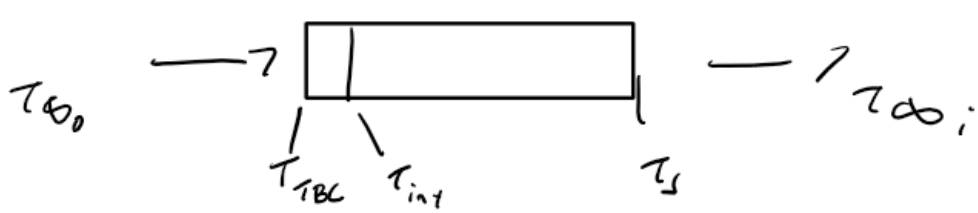
$$\boxed{T_s = 1236 \text{ K}, \quad \text{NOT below } T_{max} = 1200 \text{ K}}$$

b)



$$R_{tot} = \frac{1}{1000} + \frac{0.0005}{1} + 10^{-4} + \frac{0.005}{25} + \frac{1}{500} = 0.0038$$

$$q'' = \frac{1616 - 400}{0.0038} = 320 \text{ kW/m}^2$$



$$q'' = h_o (T_{\infty,o} - T_{TBC})$$

$$320000 = 1000 (1616 - T_{TBC})$$

$$T_{TBC} = 1296 \text{ K}$$

$$T_{TBC} - T_{int} = 320000 (0.0005)$$

$$1136 \text{ K} = T_{int}$$

$$T_{int} - T_s = 320000 (0.0001)$$

$$\boxed{\begin{array}{l} 1104 \text{ K} = T_s \\ \text{Safe, below } T_{max} = 1200 \text{K} \end{array}}$$